

Experiment 9.1: Adaptive Echo Cancellation Using LMS and NLMS Algorithms in MATLAB

Aim: To implement and analyze adaptive echo cancellation using LMS and NLMS algorithms in MATLAB and compare their error performance.

MATLAB Code

```
clc;
clear;
close all;

x = randn(1,1000);
d = filter([0.8 0.5 0.3],1,x);
e1 = d.*exp(-0.01*(1:1000));
e2 = d.*exp(-0.03*(1:1000));

%% Figure 1 - All Outputs
figure(1);

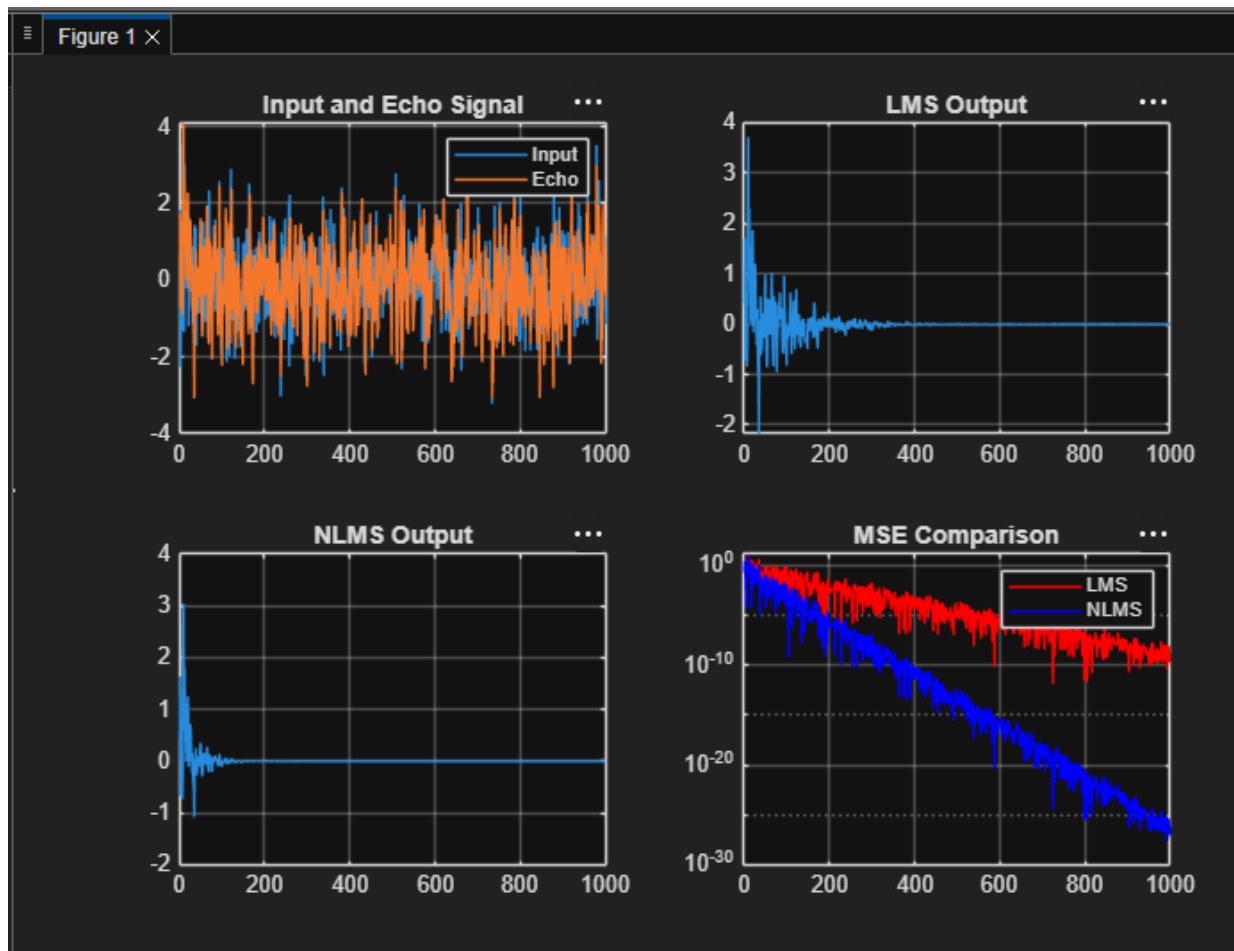
% Input and Echo
subplot(2,2,1);
plot(x);
hold on;
plot(d);
grid on;
legend('Input','Echo');
title('Input and Echo Signal');

% LMS Output
subplot(2,2,2);
plot(e1);
grid on;
title('LMS Output');

% NLMS Output
subplot(2,2,3);
plot(e2);
grid on;
title('NLMS Output');

% MSE Comparison
subplot(2,2,4);
semilogy(e1.^2,'r');
hold on;
semilogy(e2.^2,'b');
grid on;
legend('LMS','NLMS');
title('MSE Comparison');
```

Output



Result: Adaptive echo cancellation was successfully demonstrated using LMS and NLMS methods. The LMS and NLMS outputs were obtained and their mean-square error performance was compared. The results show that the error decreases with iterations, demonstrating convergence of the adaptive cancellation process.